Ajay Devidas Wanekar

M.Tech Energy Science and Technology - TA

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SUMMARY

M.Tech student at IIT Hyderabad with 2+ years of AI/ML research experience in deep learning, computer vision, and multimodal AI. Experienced in fine-tuning and benchmarking transformer-based Vision-Language Models and LLMs with PyTorch and Hugging Face, building training and evaluation pipelines for camera, near-infrared, and sonar data, with two IEEE conference publications.

EDUCATION

Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Energy Science and Technology - TA	IIT Hyderabad	2027	6.8
B.E. Electrical Engineering	Savitribai Phule Pune University	2022	7.28
XII (Maharashtra State Board)	Vasantrao Naik Jr. College of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	2017	67.38%
X (Maharashtra State Board)	Maharashtra Public School, DTP, Ch. Sambhajinagar	2014	83.00%

WORK EXPERIENCE

Project Associate | LFOVIA Lab, IIT Hyderabad **Sep 2024 – Jul 2025**

Fine-tuned and evaluated transformer-based Vision-Language Models (VLMs), including Florence-2 Base/Large, Llama 3.2 Vision, and Qwen2-VL, for multimodal text-image understanding, and benchmarked them against task-specific models including RetinaFace, DeepFace, and Siamese architectures for detection, face analysis, and visual tracking/re-identification.

Junior Research Assistant | TIHAN, IIT Hyderabad **Nov 2023 – Jul 2024**

Worked on multimodal perception for autonomous systems by developing and curating near-infrared and sonar datasets, implementing sensor-specific preprocessing, annotation, data-quality optimization, and model-training pipelines for robust object detection in complex maritime environments, resulting in an IEEE conference publication.

Research Intern | WINET Lab, IIT Hyderabad **Jul 2023 – Nov 2023**

Developed sonar-based underwater object-detection datasets by implementing annotation, Python-based preprocessing, data augmentation, and markup pipelines to improve data quality and model robustness, with the work published at the IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2023).

RESEARCH PROJECTS

Multimodal Vision-Language Model Evaluation for Computer Vision Tasks (LFOVIA Lab, IIT Hyderabad):

- Evaluated generative-AI Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision) for scene understanding, including object detection, segmentation, crowd analysis, action recognition, and text recognition, as alternatives to task-specific models.
- Benchmarked and fine-tuned models using Python-based PyTorch pipelines, evaluating detection performance (precision/recall, IoU).

Camera-Based Perception and Simulation for Autonomous Vehicles (TIHAN, IIT Hyderabad):

- Developed vision-based perception systems using YOLO and DeepSORT for real-time object detection, multi-object tracking, and position estimation of surrounding vehicles and pedestrians.
- Tested perception models in CARLA simulation across driving scenarios, and performed LiDAR and camera data annotation for training and validation.

Multi-Channel Conversational AI Platform (IIT Hyderabad Build Program):

- Built a multi-tenant generative-AI platform unifying voice, chat, and email, with a hybrid retrieval pipeline (semantic + keyword) and LLM-based intent classification that improved answer relevance by 18%.

M.TECH THESIS

Machine Learning for Non-Oxidative Methane Coupling in a DBD Reactor

- Developed and experimented with Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression models to predict hydrogen production from reactor operating conditions, validated with Leave-One-Out Cross-Validation (LOOCV).
- Recently adapted the RRSI method to this work and am experimenting with it to rigorously verify that model improvements exceed experimental noise.

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs).” IEEE i-COSTE 2023, pp. 1–5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. “Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging.” IEEE i-COSTE 2023, pp. 1–6.

TECHNICAL SKILLS

Programming & Frameworks: Python, SQL, Linux, PyTorch, TensorFlow, Hugging Face Transformers, CUDA, OpenCV, NumPy, Pandas, Scikit-learn, ROS2

Deep Learning & Generative AI: Transformer Architectures, Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision), Large Language Models (LLMs), LLM Fine-Tuning, RAG, CNN

Computer Vision & Perception: Object Detection & Multi-Object Tracking (YOLO, DeepSORT), Face Analysis (DeepFace, RetinaFace), OCR, Position Estimation, CARLA, Gazebo, Data Annotation (LiDAR, Camera, NIR, Sonar)

Research & Evaluation: Experiment Design, Model Benchmarking, Precision, Recall, IoU, Cross-Validation (LOOCV), Experimental Noise Analysis (RRSI)

HACKATHONS & ACHIEVEMENTS

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.
- AI and X Hackathon, IIT Hyderabad - 2nd Place Winner